DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

HERIOT-WATT UNIVERSITY

F78AA – Actuarial & Financial Mathematics A

December 2010

Duration: two hours

Answer ALL 7 questions

An approved calculator can be used.

Question	Marks
1	6
2	14
3	9
4	7
5	9
6	9
7	6
Total	60

1. (a) What effective rate of interest per quarter-year is equivalent to an effective rate of interest of 16% per two years? [1 mark]

(b) What effective rate of interest per 0.9 years is equivalent to an effective rate of interest of 1.3% per four months? [1 mark]

(c) Assume the effective rate of interest is 5% per annum from 1 January 2011 to 31 July 2011 and it is 0.8% per month thereafter. An investor will receive £150,000.00 on 1 May 2011 and a further £120,000.00 on 1 November 2011.

What is the total accumulated value at 31 December 2011 of these cashflows?

What is the present value of these cashflows on 1 January 2011?
[4 marks]

[Total 6 marks]

Please turn over!

2. A Financial Institution has agreed to provide a mortgage of £100,000 to an individual. The terms of the mortgage require the individual to make level annual instalments in arrears for 25 years, at an effective rate of interest of 5% p.a., to completely repay the mortgage.

(a) Determine the individual's level annual mortgage repayment.

[3 marks]

(b) Denoting the annual repayment by X, by setting out the loan schedule for payments in the first t years, or otherwise, derive the formulae for the interest content and the capital content of the payment at time t and the capital outstanding immediately after this payment is made.

[4 marks]

(c) Immediately after the 8th payment has been made, the individual decides to return to university for 4 years to study for an Honours Degree in Actuarial Science. The lender agrees with the individual that he can suspend payments for the duration of the course although interest will still accrue on the outstanding loan. Calculate the new increased level annual payment from years 13 to 25 when he returns to employment.

[4 marks]

(d) Immediately after payment is made at the end of the 15th year, he qualifies as an actuary. The lender agrees that the overall limit on his borrowing with that institution can now be £200,000. By calculating the outstanding capital to be repaid under the original loan agreement, what is the maximum incremental loan now available to the individual.

[3 marks]

[Total 14 marks]

Please turn over!

3. Consider the following investment fund.

time in years	0	0.75	1.6	3	5
payments into the fund	169		5000		
payments out of the fund		37		5100	
fund value just before payment	0	163	105	5200	170

(a) Calculate the Time-Weighted Rate of Return (TWRR).

[4 marks]

(b) Show that the Money-Weighted Rate of Return (MWRR) is approximately 1.77% p.a..

[3 marks]

(c) Briefly explain the difference between the TWRR and the MWRR using the results from (a) and (b) to illustrate the points you make.

[2 marks]

[Total 9 marks]

Please turn over!

4. (a) Explain briefly the important features of an index-linked gilt **and** describe the attractions of an indexed linked gilt to investors.

[2 marks]

(b) On 15 July 2010 the government of a country issued an index-linked gilt with a nominal coupon rate of 3% per annum payable half-yearly. Coupons will be paid on 15 July and 15 January each year. The first coupon payment is due on 15 January 2011, the last coupon is due on 15 July 2035. The bond will be redeemed on 15 July 2035. The nominal redemption rate is 100%. All coupons and the redemption payment will be adjusted according to the value of the inflation index of that country three months prior to the payment. On 15 July 2010 the last known value of the inflation index Q was Q(April 2010). This is also the base value of the inflation index for the adjustment of coupon payments and the redemption amount.

What payments will an investor receive per £100 nominal of the stock on 15 January 2013 and on 15 July 2035 if the actual annual rate of inflation between April 2010 and July 2035 is constant and equal to 1.9% per annum?

[4 marks]

(c) An investor is interested in a certain **monetary yield**. He wants to invest into government bonds. Should the investor invest into index-linked gilts or into government bonds which are not index-linked?

[1 mark]

[Total 7 marks]

Please turn over!

5. On 10 August 2009 a company issued a number of bonds, each of nominal amount £100.00, to be redeemed at a redemption rate of 115% on 10 August 2024. The coupon rate is 8% per annum, payable quarter-yearly on each 10 November, 10 February, 10 May and 10 August. The first coupon payment was due on 10 November 2009, and the last will be due on 10 August 2024.

(a) On 10 August 2009 the bond was priced such that an investor who is liable to income tax at a rate of 40% and capital gains tax at a rate of 18% obtains a redemption yield of 6.975% per annum after tax. What was the issue price per £100.00 nominal of the bond?

[5 marks]

(b) A different investor who is also liable to capital gains tax at a rate of 18% bought the bond at the issue date for the price calculated in part (a). This investor claims that his yield is 2% per quarter-year as he pays less income tax than the investor in part (a). Calculate the income tax rate of the new investor.

[4 marks]

[Total 9 marks]

Please turn over!

6. (a) If i is the effective rate of interest per annum, starting from first principles show that:-

$$a^{i}_{\overline{n}|} = \frac{1 - v^n}{i}$$

where $v = \frac{1}{(1+i)}$ and hence explain by general reasoning that

$$s^{i}_{\overline{n}|} = (1+i)^n a^{i}_{\overline{n}|} \, .$$

[4 marks]

(b) Show that the present value of an annuity which pays 1 p.a. in arrears, only in **even** years j, where ($j = 2, 4, 6, \ldots, n, n$ is even) denoted by $a^{i}_{\overline{n,even}|}$ is given by the expression:-

$$a^{i}_{\overline{n,even}|} = \frac{1}{(i+2)} a^{i}_{\overline{n}|} \, .$$

[3 marks]

(c) Hence, or otherwise, show that $s^{i}_{\overline{n,odd}|}$, the accumulation of an annuity at time n which pays 1 p.a. in arrears only in **odd** years j, where ($j = 1, 3, 5, \ldots, n-1, n$ is even) is given by the expression:-

$$s^{i}_{\overline{n,odd}|} = \frac{(1+i)}{(i+2)} s^{i}_{\overline{n}|} \, .$$

[2 marks]

[Total 9 marks]

Please turn over!

7. A **Payday Loan** is defined as a small loan over a short time period that is intended to cover the borrower's expenses until his or her next pay date. To cover the fixed expenses of providing this loan an administration charge is added to each loan **to be repaid at the end of the loan period** regardless of the size of the loan provided.

An unfortunate individual requires to borrow £100 for 7 days until the individual's salary is paid. He compares the terms of two lenders A and B, as follows:-

Lender A Administration charge of £12.50, with **simple** interest calculated as 0.75% per day on the sum of the initial loan and the administrative charge, and

Lender B Administration charge of £15.00, with **compound** interest calculated as 0.25% per day on **only** the initial loan.

(a) Showing all your working, which lender should the individual choose?

[3 marks]

(b) Based on this lender's terms, and assuming that there are 365 days in the year, calculate the effective annualised rate of interest the borrower is charged.

[3 marks]

[Total 6 marks]

END OF PAPER